Lesson 61: Stochastic Game Theory

AI/ML Mathematics

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1 Motivation and Overview

Consider a fleet of autonomous vehicles approaching a busy intersection without traffic lights. Each vehicle is an independent agent choosing an action—accelerate, brake, or turn—based on the current configuration of all vehicles (the state). The outcome each vehicle experiences depends not only on its own action but on the *joint* actions of every other vehicle. After each time step, the vehicles transition to a new configuration, and the process repeats. The classical MDP framework from Lessons 39–46 cannot capture this setting: an MDP has a single decision-maker, whereas here multiple agents with potentially conflicting objectives interact over time. How should we formalize sequential multi-agent interaction? Under what conditions do equilibria exist? How can agents compute or learn them?

The mathematical framework that answers these questions is the *stochastic game*, introduced by Shapley [1953]. A stochastic game extends an MDP to multiple players, and extends a static game to multiple stages with state transitions. The tools developed in this lesson will let us define the stochastic game formalism precisely, prove the existence of equilibrium values in zero-sum stochastic games via Shapley's theorem, establish the existence of Markov perfect equilibria in general-sum games, and understand the computational barriers to finding equilibria.

Why Stochastic Games Matter for RL/ML

- **Minimax-Q learning.** The Q-learning extension to zero-sum stochastic games directly uses the Shapley operator and minimax value characterization developed in this lesson.
- **Nash-Q learning.** General-sum MARL algorithms such as Nash-Q rely on the Markov perfect equilibrium concept and the multi-agent Bellman equations defined here.
- **Self-play (AlphaGo/AlphaZero).** Self-play training in two-player zero-sum games converges to the game value characterized by Shapley's theorem.
- **MADDPG and centralized training.** Multi-agent policy gradient methods operate in the stochastic game framework, using the joint-action Bellman equations as their theoretical backbone.

Prerequisites. Markov decision processes: Bellman equations, value iteration, policy iteration, contraction mappings (Lessons 39–46). Normal-form games, Nash equilibrium, minimax theorem, Kakutani's fixed-point theorem (Lessons 20–21 from Phase 01 game theory). Banach fixed-point theorem and properties of contraction mappings on complete metric spaces (Lessons 15–16 from Phase 01 functional analysis).

Learning objectives. After completing this lesson, you will be able to:

- **Define** a stochastic game and **classify** its special cases (zero-sum, team, potential).
- **Derive** the multi-agent Bellman equations for the value function under a joint policy profile.
- **Prove** Shapley's theorem on the existence of a value and optimal stationary strategies in zero-sum stochastic games.
- **State** and **explain** the Fink–Takahashi theorem on the existence of Markov perfect equilibria in general-sum stochastic games.
- **Prove** that multiplicative weights achieve sublinear external regret, and **explain** how no-regret learning connects to coarse correlated equilibrium.
- **Analyze** the computational complexity of equilibrium computation, including the PPAD-completeness barrier.

2 References

- Shoham and Leyton-Brown [2008, Chapters 3–4] — Review of normal-form and extensive-form games; Chapters 6–7 cover learning in games.
- Filar and Vrieze [1997, Chapters 1–5] — Comprehensive treatment of competitive Markov decision processes (stochastic games), including Shapley's value theorem and equilibrium existence.
- Fudenberg and Tirole [1991, Chapters 1–3] — Static games, Nash equilibrium, dynamic games, subgame perfection, and repeated games.
- Zhang et al. [2021] — Modern survey of multi-agent reinforcement learning theory and algorithms, providing context for the stochastic game framework.
- Shapley [1953] — The original paper introducing stochastic games and proving the minimax theorem for discounted zero-sum stochastic games.

3 Normal-Form Games

We begin by reviewing the static game concepts that form the single-stage building blocks of stochastic games. Readers who have completed Lessons 20–21 may skim this section; however, the notation established here is used throughout the remainder of the lesson.

Definition 3.1 (Normal-Form Game)**.** A *finite normal-form game* is a tuple $G = (\mathcal{N}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \{u_i\}_{i\in\mathcal{N}})$ where:

1. $\mathcal{N} = \{1, \dots, n\}$ is the finite set of players;
2. $\mathcal{A}_i$ is the finite action set of player i;
3. $u_i \colon \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \mathbb{R}$ is the utility (payoff) function of player i.

We write $\mathbf{a} = (a_1, \dots, a_n) \in \boldsymbol{\mathcal{A}} := \mathcal{A}_1 \times \cdots \times \mathcal{A}_n$ for a joint action profile, and $a_{-i} = (a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n)$ for the actions of all players except i.

Definition 3.2 (Mixed Strategy)**.** A *mixed strategy* for player i is a probability distribution $\sigma_i \in \Delta(\mathcal{A}_i)$, where $\Delta(\mathcal{A}_i) = \{\sigma_i \in \mathbb{R}^{|\mathcal{A}_i|}_{\geq 0} : \sum_{a_i\in\mathcal{A}_i} \sigma_i(a_i) = 1\}$ is the probability simplex. A *mixed strategy profile* is $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n) \in \prod_{i=1}^n \Delta(\mathcal{A}_i)$. Under independent mixing, the expected payoff to player i is:

$$u_i(\boldsymbol{\sigma}) = \sum_{\mathbf{a}\in\boldsymbol{\mathcal{A}}} \Big(\prod_{j=1}^n \sigma_j(a_j)\Big) u_i(\mathbf{a}). \tag{1}$$

Definition 3.3 (Best Response)**.** The *best response correspondence* of player i to the opponents' profile σ_{-i} is:

$$\mathrm{BR}_i(\sigma_{-i}) = \arg\max_{\sigma_i\in\Delta(\mathcal{A}_i)} u_i(\sigma_i, \sigma_{-i}). \tag{2}$$

Since u_i is linear in σ_i, the set $\mathrm{BR}_i(\sigma_{-i})$ is always a nonempty convex polytope (a face of $\Delta(\mathcal{A}_i)$).

Definition 3.4 (Nash Equilibrium)**.** A mixed strategy profile $\boldsymbol{\sigma}^* = (\sigma_1^*, \dots, \sigma_n^*)$ is a *Nash equilibrium* (NE) if for every player $i \in \mathcal{N}$:

$$\sigma_i^* \in \mathrm{BR}_i(\sigma_{-i}^*), \quad \text{equivalently} \quad u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*) \quad \forall\, \sigma_i \in \Delta(\mathcal{A}_i). \tag{3}$$

Theorem 3.5 (Nash's Existence Theorem [Nash, 1950, 1951])**.** *Every finite normal-form game has at least one Nash equilibrium in mixed strategies.*

Proof. Define the joint strategy space $\Sigma = \prod_{i=1}^{n} \Delta(\mathcal{A}_i)$. This is a nonempty, compact, convex subset of $\mathbb{R}^{\sum_i |\mathcal{A}_i|}$. Define the correspondence $\mathrm{BR}\colon \Sigma \rightrightarrows \Sigma$ by $\mathrm{BR}(\boldsymbol{\sigma}) = \prod_{i=1}^{n} \mathrm{BR}_i(\sigma_{-i})$. We verify the hypotheses of Kakutani's fixed-point theorem:

1. Σ is a nonempty, compact, convex subset of a finite-dimensional Euclidean space.
2. For each $\boldsymbol{\sigma}$, $\mathrm{BR}(\boldsymbol{\sigma})$ is nonempty (since each $\mathrm{BR}_i(\sigma_{-i})$ is nonempty, as a continuous function on a compact set attains its maximum) and convex (since each $\mathrm{BR}_i(\sigma_{-i})$ is a face of the simplex, hence convex, and the product of convex sets is convex).
3. BR has a closed graph. Suppose $\boldsymbol{\sigma}^{(k)} \to \boldsymbol{\sigma}$ and $\boldsymbol{\tau}^{(k)} \to \boldsymbol{\tau}$ with $\boldsymbol{\tau}^{(k)} \in \mathrm{BR}(\boldsymbol{\sigma}^{(k)})$ for all k. Then for each i and any $\sigma_i' \in \Delta(\mathcal{A}_i)$, $u_i(\tau_i^{(k)}, \sigma_{-i}^{(k)}) \geq u_i(\sigma_i', \sigma_{-i}^{(k)})$. Taking $k \to \infty$ and using continuity of u_i: $u_i(\tau_i, \sigma_{-i}) \geq u_i(\sigma_i', \sigma_{-i})$, so $\tau_i \in \mathrm{BR}_i(\sigma_{-i})$, hence $\boldsymbol{\tau} \in \mathrm{BR}(\boldsymbol{\sigma})$.

By Kakutani's fixed-point theorem (Lesson 16, Theorem 5.4), BR has a fixed point $\boldsymbol{\sigma}^*$, i.e., $\boldsymbol{\sigma}^* \in \mathrm{BR}(\boldsymbol{\sigma}^*)$. This is precisely a Nash equilibrium. □

Example 3.6 (Prisoner's Dilemma)**.** Two players each choose Cooperate (C) or Defect (D). Payoffs:

	C	D
C	$(3,3)$	$(0,5)$
D	$(5,0)$	$(1,1)$

Regardless of the opponent's action, D strictly dominates C for both players: $u_i(D, a_{-i}) > u_i(C, a_{-i})$ for all a_{-i}. Therefore the unique Nash equilibrium is (D, D) with payoffs $(1,1)$—even though (C, C) with payoffs $(3,3)$ Pareto-dominates it.

Example 3.7 (Matching Pennies)**.** Two players simultaneously choose Heads (H) or Tails (T):

	H	T
H	$(1,-1)$	$(-1,1)$
T	$(-1,1)$	$(1,-1)$

This is a zero-sum game. There is no pure-strategy Nash equilibrium. The unique mixed equilibrium is $\sigma_1 = \sigma_2 = (\frac{1}{2}, \frac{1}{2})$ with value 0.

We now turn to the dynamic extensions of normal-form games, beginning with repeated games.

4 Repeated Games and Fictitious Play

When a normal-form game is played repeatedly, new equilibria emerge that exploit the temporal structure. Repeated games bridge the gap between static games and the fully general stochastic game model.

Definition 4.1 (Repeated Game)**.** Let $G = (\mathcal{N}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \{u_i\}_{i\in\mathcal{N}})$ be a finite normal-form game (the *stage game*). The *T-fold repeated game* G^T consists of playing G for T rounds. At each round $t = 1, \ldots, T$, all players simultaneously choose actions, and the joint action $\mathbf{a}^t$ is revealed. A (behavioral) strategy for player i is a mapping from histories $h^t = (\mathbf{a}^1, \ldots, \mathbf{a}^{t-1})$ to $\Delta(\mathcal{A}_i)$. The *infinitely repeated game* G^∞ extends this to $T = \infty$ with discounted payoff $U_i = (1-\gamma)\sum_{t=1}^{\infty} \gamma^{t-1} u_i(\mathbf{a}^t)$, where $\gamma \in (0,1)$ is the common discount factor.

Theorem 4.2 (Folk Theorem (Informal))**.** *In the infinitely repeated game G^∞ with sufficiently patient players (γ close to 1), any feasible payoff vector that strictly dominates the minimax payoff vector can be sustained as a Nash equilibrium of the repeated game.*

Remark 4.3. The folk theorem implies that repetition vastly expands the set of equilibrium outcomes beyond the static Nash equilibria. A rigorous proof relies on constructing punishment strategies that deter deviation; see Fudenberg and Tirole [1991, Chapter 5] for the formal statement and proof. For MARL, the folk theorem warns us that convergence to a *particular* equilibrium requires additional structure (e.g., no-regret guarantees or coordination mechanisms).

Definition 4.4 (Fictitious Play [Brown, 1951])**.** In *fictitious play*, each player maintains an empirical frequency of opponents' past actions and best-responds to it. Formally, let $n_i^t(a_i)$ be the number of times player i has played action a_i in rounds $1, \ldots, t$, and define the empirical distribution $\hat{\sigma}_i^t(a_i) = n_i^t(a_i)/t$. At round $t+1$, player i plays any action in $\mathrm{BR}_i(\hat{\sigma}_{-i}^t)$.

Theorem 4.5 (Robinson's Convergence Theorem [Robinson, 1951])**.** *In any two-player zero-sum game, fictitious play converges: the empirical frequencies $(\hat{\sigma}_1^t, \hat{\sigma}_2^t)$ converge to the set of Nash equilibria as $t \to \infty$.*

Proof. Let $v^* = \max_{\sigma_1} \min_{\sigma_2} u_1(\sigma_1, \sigma_2)$ be the value of the game (which exists by the minimax theorem, Lesson 20, Theorem 4.1). Define the "regret" quantities:

$$\overline{v}^t = \frac{1}{t} \sum_{\tau=1}^{t} \max_{a_1 \in \mathcal{A}_1} u_1(a_1, a_2^\tau), \quad \underline{v}^t = \frac{1}{t} \sum_{\tau=1}^{t} \min_{a_2 \in \mathcal{A}_2} u_1(a_1^\tau, a_2). \tag{4}$$

Since player 1 best-responds to $\hat{\sigma}_2^{t-1}$, we have $u_1(a_1^t, \hat{\sigma}_2^{t-1}) = \max_{a_1} u_1(a_1, \hat{\sigma}_2^{t-1})$, which gives $\overline{v}^t \geq \underline{v}^t$ for all t. Moreover, $\overline{v}^t \geq v^*$ and $\underline{v}^t \leq v^*$ hold by the minimax inequality. Robinson's argument proceeds by showing that the gap $\overline{v}^t - \underline{v}^t$ is $O(1/\sqrt{t})$: since payoffs are bounded in $[-M, M]$, the increment from round t to $t+1$ in each quantity is at most $O(M/t)$. By a careful accounting of the increments (see Robinson 1951 for the complete inductive argument), one establishes that both $\overline{v}^t$ and $\underline{v}^t$ converge to v^*.

It follows that $u_1(\hat{\sigma}_1^t, \hat{\sigma}_2^t) \to v^*$. Any limit point (σ_1^*, σ_2^*) of the empirical frequencies satisfies the minimax conditions and is therefore a Nash equilibrium. □

Example 4.6 (Fictitious Play in Matching Pennies)**.** In the matching pennies game (Example 3.7), suppose both players initially play H. Player 1 (who wants matching) has no incentive to deviate. Player 2 (who wants mismatching) switches to T at round 2. Now player 1's empirical model of player 2 is $(\frac{1}{2}, \frac{1}{2})$, so player 1 is indifferent and may stay with H. Player 2 now sees player 1 always playing H, so player 2 continues with T. As rounds progress, the empirical frequencies oscillate but converge to the unique equilibrium $(\frac{1}{2}, \frac{1}{2})$, consistent with Robinson's theorem.

Warning

Fictitious play does *not* converge in all games. Shapley [1953] (not to be confused with his stochastic games paper) exhibited a 3×3 game in which fictitious play cycles and the empirical frequencies do not converge to any Nash equilibrium. Convergence is guaranteed for zero-sum games (Robinson), potential games, and $2 \times n$ games, but not in general.

With the single-stage and repeated-game foundations in place, we turn to no-regret learning, which provides equilibrium convergence guarantees without assuming opponent rationality.

5 No-Regret Learning

No-regret learning offers a fundamentally different path to equilibrium: instead of assuming opponents play rationally, each agent independently minimizes its own regret, and equilibrium emerges as a consequence. This perspective is central to modern MARL theory.

Definition 5.1 (External Regret)**.** Consider player i choosing actions $a_i^1, \ldots, a_i^T$ over T rounds while observing opponent actions $a_{-i}^1, \ldots, a_{-i}^T$. The *external regret* of player i after T rounds is:

$$\text{Regret}_T^i = \max_{a_i \in \mathcal{A}_i} \sum_{t=1}^{T} u_i(a_i, a_{-i}^t) - \sum_{t=1}^{T} u_i(a_i^t, a_{-i}^t). \tag{5}$$

An algorithm is *no-regret* (or *Hannan consistent*) if $\text{Regret}_T^i / T \to 0$ as $T \to \infty$.

Definition 5.2 (Multiplicative Weights Update (MWU))**.** Let $\eta > 0$ be a learning rate and assume utilities are bounded: $u_i(a_i, a_{-i}) \in [0, 1]$ for all a_i, a_{-i}. Initialize weights $w_i^1(a_i) = 1$ for all $a_i \in \mathcal{A}_i$. At each round t:

1. Play action a_i with probability $\sigma_i^t(a_i) = w_i^t(a_i)/W_i^t$ where $W_i^t = \sum_{a_i'} w_i^t(a_i')$.
2. Observe opponent actions a_{-i}^t (or, in the bandit variant, only the payoff $u_i(a_i^t, a_{-i}^t)$).
3. Update weights: $w_i^{t+1}(a_i) = w_i^t(a_i) \cdot \exp(\eta\, u_i(a_i, a_{-i}^t))$.

Theorem 5.3 (MWU Regret Bound [Freund and Schapire, 1999])**.** *With learning rate* $\eta = \sqrt{\ln |\mathcal{A}_i|/T}$ *and utilities in* $[0, 1]$*, the multiplicative weights update satisfies:*

$$\text{Regret}_T^i \leq 2\sqrt{T \ln |\mathcal{A}_i|}. \tag{6}$$

Proof. Define the potential $\Phi^t = \ln W_i^t$. We have $\Phi^1 = \ln |\mathcal{A}_i|$ (since all weights start at 1). At round t:

$$\begin{aligned} \Phi^{t+1} - \Phi^t = \ln \frac{W_i^{t+1}}{W_i^t} &= \ln \frac{\sum_{a_i} w_i^t(a_i) \exp(\eta\, u_i(a_i, a_{-i}^t))}{W_i^t} \\ &= \ln \sum_{a_i} \sigma_i^t(a_i) \exp(\eta\, u_i(a_i, a_{-i}^t)). \end{aligned} \tag{7}$$

Using the inequality $e^x \leq 1 + x + x^2$ for $x \in [0, 1]$ (valid when $\eta \leq 1$ and $u_i \in [0, 1]$, so that $\eta u_i \leq 1$), we obtain:

$$\Phi^{t+1} - \Phi^t \leq \ln\Big(1 + \eta \sum_{a_i} \sigma_i^t(a_i) u_i(a_i, a_{-i}^t) + \eta^2 \sum_{a_i} \sigma_i^t(a_i) u_i(a_i, a_{-i}^t)^2\Big). \tag{8}$$

Since $\ln(1 + x) \leq x$ for all $x > -1$, this gives:

$$\Phi^{t+1} - \Phi^t \leq \eta\, u_i(\sigma_i^t, a_{-i}^t) + \eta^2 \sum_{a_i} \sigma_i^t(a_i) u_i(a_i, a_{-i}^t)^2 \leq \eta\, u_i(\sigma_i^t, a_{-i}^t) + \eta^2, \tag{9}$$

where the last inequality uses $u_i(\cdot)^2 \leq 1$. Summing over $t = 1, \ldots, T$:

$$\Phi^{T+1} \leq \ln |\mathcal{A}_i| + \eta \sum_{t=1}^{T} u_i(\sigma_i^t, a_{-i}^t) + \eta^2 T. \tag{10}$$

Now we lower-bound Φ^{T+1}. For any fixed action $a_i^* \in \mathcal{A}_i$:

$$\Phi^{T+1} = \ln W_i^{T+1} \geq \ln w_i^{T+1}(a_i^*) = \eta \sum_{t=1}^{T} u_i(a_i^*, a_{-i}^t). \tag{11}$$

Combining with (10):

$$\eta \sum_{t=1}^{T} u_i(a_i^*, a_{-i}^t) \leq \ln |\mathcal{A}_i| + \eta \sum_{t=1}^{T} u_i(\sigma_i^t, a_{-i}^t) + \eta^2 T. \tag{12}$$

Rearranging and maximizing over a_i^*:

$$\text{Regret}_T^i \leq \frac{\ln |\mathcal{A}_i|}{\eta} + \eta T. \tag{13}$$

Setting $\eta = \sqrt{\ln |\mathcal{A}_i|/T}$ minimizes the right-hand side:

$$\text{Regret}_T^i \leq 2\sqrt{T \ln |\mathcal{A}_i|}. \tag{14}$$

□

Definition 5.4 (Coarse Correlated Equilibrium)**.** A distribution $\mu \in \Delta(\boldsymbol{\mathcal{A}})$ is a *coarse correlated equilibrium* (CCE) if for every player i and every action $a_i' \in \mathcal{A}_i$:

$$\sum_{\mathbf{a} \in \boldsymbol{\mathcal{A}}} \mu(\mathbf{a})\, u_i(\mathbf{a}) \geq \sum_{\mathbf{a} \in \boldsymbol{\mathcal{A}}} \mu(\mathbf{a})\, u_i(a_i', a_{-i}). \tag{15}$$

Equivalently, no player can improve their expected payoff by committing to a fixed action *before* the joint action is drawn from μ.

Theorem 5.5 (No-Regret Implies CCE Convergence)**.** *Suppose all n players use no-regret algorithms. Define the empirical distribution of joint play: $\hat{\mu}^T(\mathbf{a}) = \frac{1}{T}\sum_{t=1}^T \mathbf{1}[\mathbf{a}^t = \mathbf{a}]$. Then every limit point of $\{\hat{\mu}^T\}_{T \geq 1}$ is a coarse correlated equilibrium of the stage game.*

Proof. Fix player i and any action $a_i' \in \mathcal{A}_i$. The no-regret condition gives:

$$\sum_{t=1}^T u_i(a_i', a_{-i}^t) - \sum_{t=1}^T u_i(a_i^t, a_{-i}^t) \leq \text{Regret}_T^i = o(T). \tag{16}$$

Dividing by T:

$$\frac{1}{T}\sum_{t=1}^T u_i(a_i', a_{-i}^t) - \frac{1}{T}\sum_{t=1}^T u_i(\mathbf{a}^t) \leq \frac{\text{Regret}_T^i}{T} \to 0. \tag{17}$$

Rewriting in terms of the empirical distribution:

$$\sum_{\mathbf{a}} \hat{\mu}^T(\mathbf{a})\, u_i(a_i', a_{-i}) - \sum_{\mathbf{a}} \hat{\mu}^T(\mathbf{a})\, u_i(\mathbf{a}) \leq \frac{\text{Regret}_T^i}{T}. \tag{18}$$

At any limit point μ^* of $\hat{\mu}^T$, the right side vanishes, giving the CCE condition (15) for player i and action a_i'. Since i and a_i' were arbitrary, μ^* is a CCE. □

No-Regret to Equilibrium

If every player in a finite game independently runs a no-regret algorithm (such as multiplicative weights), the empirical joint distribution of play converges to the set of coarse correlated equilibria. This requires no coordination between players and no knowledge of opponents' strategies—only that each player observes the utilities it receives. The convergence rate is $O(\sqrt{\ln |\mathcal{A}_i|/T})$ in average regret when using MWU.

Having established both classical (fictitious play) and modern (no-regret) approaches to learning in repeated games, we now generalize to the full stochastic game model.

6 Stochastic Game Formalism

A stochastic game combines the strategic interaction of a normal-form game with the state dynamics of a Markov decision process.

Definition 6.1 (Stochastic Game)**.** A *(discounted) stochastic game* is a tuple $\Gamma = (\mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \mathcal{P}, \{R_i\}_{i\in\mathcal{N}}, \gamma$ where:

1. $\mathcal{N} = \{1, \dots, n\}$ is the finite set of players;
2. $\mathcal{S}$ is the finite state space;
3. $\mathcal{A}_i$ is the finite action set of player i; write $\boldsymbol{\mathcal{A}} = \prod_{i=1}^n \mathcal{A}_i$ for the joint action space;
4. $\mathcal{P}\colon \mathcal{S} \times \boldsymbol{\mathcal{A}} \to \Delta(\mathcal{S})$ is the transition kernel: $\mathcal{P}(s' \mid s, \mathbf{a})$ is the probability of transitioning to state s' given state s and joint action $\mathbf{a}$;
5. $R_i\colon \mathcal{S} \times \boldsymbol{\mathcal{A}} \to \mathbb{R}$ is the reward function of player i;
6. $\gamma \in [0, 1)$ is the common discount factor.

Remark 6.2. When $n = 1$, the stochastic game reduces to an MDP (Lesson 39). When $|\mathcal{S}| = 1$, it reduces to a repeated game (Definition 4.1). Stochastic games thus unify both frameworks.

Definition 6.3 (Markov Strategy)**.** A *(stationary) Markov strategy* for player i is a mapping $\pi_i\colon \mathcal{S} \to \Delta(\mathcal{A}_i)$, where $\pi_i(a_i \mid s)$ denotes the probability that player i selects action a_i in state s. A *joint Markov strategy profile* is $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$. More generally, a *behavior strategy* may depend on the full history $(s^1, \mathbf{a}^1, s^2, \mathbf{a}^2, \dots, s^t)$; however, for the discounted criterion, it suffices to consider Markov strategies (as we shall see).

Definition 6.4 (Value Function in Stochastic Games)**.** Given a joint strategy profile $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$, the *value function* of player i starting from state s is:

$$V_i^{\boldsymbol{\pi}}(s) = \mathbb{E}^{\boldsymbol{\pi}}\left[\sum_{t=0}^{\infty} \gamma^t R_i(s^t, \mathbf{a}^t) \;\middle|\; s^0 = s\right], \tag{19}$$

where the expectation is over the trajectory induced by the joint policy and the transition kernel.

Theorem 6.5 (Multi-Agent Bellman Equation)**.** *Under any stationary joint policy $\boldsymbol{\pi}$, the value function $V_i^{\boldsymbol{\pi}}$ satisfies the Bellman equation: for all $s \in \mathcal{S}$,*

$$V_i^{\boldsymbol{\pi}}(s) = \sum_{\mathbf{a}\in\boldsymbol{\mathcal{A}}} \left(\prod_{j=1}^n \pi_j(a_j \mid s)\right) \left[R_i(s, \mathbf{a}) + \gamma \sum_{s'\in\mathcal{S}} \mathcal{P}(s' \mid s, \mathbf{a})\, V_i^{\boldsymbol{\pi}}(s')\right]. \tag{20}$$

Proof. This follows from the same argument as the single-agent case (Lesson 41, Theorem 3.1). Condition on the first-step joint action $\mathbf{a}$:

$$\begin{aligned} V_i^{\boldsymbol{\pi}}(s) &= \mathbb{E}^{\boldsymbol{\pi}}\left[R_i(s^0, \mathbf{a}^0) + \gamma V_i^{\boldsymbol{\pi}}(s^1) \mid s^0 = s\right] && (21)\\ &= \sum_{\mathbf{a}} \mathbb{P}^{\boldsymbol{\pi}}(\mathbf{a}^0 = \mathbf{a} \mid s^0 = s)\left[R_i(s, \mathbf{a}) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, \mathbf{a}) V_i^{\boldsymbol{\pi}}(s')\right]. && (22) \end{aligned}$$

Under independent stationary strategies, $\mathbb{P}^{\boldsymbol{\pi}}(\mathbf{a}^0 = \mathbf{a} \mid s^0 = s) = \prod_{j=1}^n \pi_j(a_j \mid s)$, which yields (20). □

From MDP Bellman to Multi-Agent Bellman

Equation (20) generalizes the single-agent Bellman equation $V^\pi(s) = \sum_a \pi(a|s)[R(s,a) + \gamma \sum_{s'} \mathcal{P}(s'|s,a)V^\pi(s')]$ from Lesson 41. The key difference is the product $\prod_j \pi_j(a_j|s)$ reflecting independent strategy choices by multiple agents. All value-based MARL algorithms (Minimax-Q, Nash-Q, Friend/Foe-Q) build on this equation.

Example 6.6 (Two-Player Grid World Stochastic Game)**.** Consider a 2×2 grid with states $\mathcal{S} = \{s_1, s_2, s_3, s_4\}$ and two players. Player 1 chooses $\{L, R\}$ (left or right) and Player 2 chooses $\{U, D\}$ (up or down). In state s_1, the rewards are:

$R_1(s_1, \cdot)$	U	D
L	2	-1
R	0	1

$R_2(s_1, \cdot)$	U	D
L	-2	1
R	0	-1

The transition kernel maps $(s_1, L, U) \to s_2$ deterministically, $(s_1, R, D) \to s_3$, etc. Each state has its own reward matrix and transitions, making this a stochastic game with $|\mathcal{S}| = 4$, $|\mathcal{A}_1| = |\mathcal{A}_2| = 2$. The multi-agent Bellman equation (20) couples the values across all four states.

We now specialize to the zero-sum case, where the game-theoretic and dynamic programming perspectives yield the strongest results.

7 Zero-Sum Stochastic Games

Definition 7.1 (Zero-Sum Stochastic Game)**.** A two-player stochastic game is *zero-sum* if $R_1(s, \mathbf{a}) = -R_2(s, \mathbf{a})$ for all $(s, \mathbf{a}) \in \mathcal{S} \times \boldsymbol{\mathcal{A}}$. We write $R(s, a_1, a_2) := R_1(s, a_1, a_2)$ for the payoff to Player 1 (the maximizer), and $-R$ for the payoff to Player 2 (the minimizer).

In a zero-sum stochastic game, we seek the *value* of the game and the optimal strategies for both players.

Definition 7.2 (Minimax Value)**.** The *(upper) minimax value* of a zero-sum stochastic game starting from state s is:

$$\overline{V}(s) = \inf_{\pi_2} \sup_{\pi_1} V_1^{(\pi_1, \pi_2)}(s), \tag{23}$$

and the *lower minimax value* is:

$$\underline{V}(s) = \sup_{\pi_1} \inf_{\pi_2} V_1^{(\pi_1, \pi_2)}(s). \tag{24}$$

The game *has a value* if $\overline{V}(s) = \underline{V}(s) =: V^*(s)$ for all s.

Definition 7.3 (Shapley Operator)**.** For a zero-sum stochastic game, the *Shapley operator* $\mathcal{H} \colon \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ is defined for each $s \in \mathcal{S}$ by:

$$(\mathcal{H}V)(s) = \max_{\sigma_1 \in \Delta(\mathcal{A}_1)} \min_{\sigma_2 \in \Delta(\mathcal{A}_2)} \sum_{a_1, a_2} \sigma_1(a_1)\sigma_2(a_2) \Big[R(s, a_1, a_2) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a_1, a_2) V(s') \Big]. \tag{25}$$

By the minimax theorem for finite zero-sum games (Lesson 20, Theorem 4.1), the $\max\min$ equals the $\min\max$, so $\mathcal{H}$ is well-defined and we may equivalently write $\min\max$ in (25).

Lemma 7.4 (Shapley Operator is a Contraction)**.** *The Shapley operator $\mathcal{H}$ is a γ-contraction in the supremum norm: for all $V, W \in \mathbb{R}^{|\mathcal{S}|}$,*

$$\|\mathcal{H}V - \mathcal{H}W\|_\infty \le \gamma \|V - W\|_\infty. \tag{26}$$

Proof. Fix $s \in \mathcal{S}$. Let (σ_1^V, σ_2^V) be the minimax strategies achieving $(\mathcal{H}V)(s)$, and similarly (σ_1^W, σ_2^W) for $(\mathcal{H}W)(s)$. Define $Q(s, a_1, a_2; V) = R(s, a_1, a_2) + \gamma \sum_{s'} \mathcal{P}(s'|s, a_1, a_2) V(s')$ and similarly $Q(s, a_1, a_2; W)$.

Note that for any σ_1, σ_2:

$$\big| \mathbb{E}_{\sigma_1, \sigma_2}[Q(s, \cdot, \cdot; V)] - \mathbb{E}_{\sigma_1, \sigma_2}[Q(s, \cdot, \cdot; W)] \big| \le \gamma \|V - W\|_\infty, \tag{27}$$

since the reward terms cancel and the transition probabilities sum to 1.

For the upper bound on $(\mathcal{H}V)(s) - (\mathcal{H}W)(s)$:

$$(\mathcal{H}V)(s) = \max_{\sigma_1} \min_{\sigma_2} \mathbb{E}_{\sigma_1,\sigma_2}[Q(s,\cdot,\cdot;V)] \tag{28}$$

$$\leq \max_{\sigma_1} \min_{\sigma_2} \big(\mathbb{E}_{\sigma_1,\sigma_2}[Q(s,\cdot,\cdot;W)] + \gamma\|V-W\|_\infty\big) \tag{29}$$

$$= (\mathcal{H}W)(s) + \gamma\|V-W\|_\infty. \tag{30}$$

By symmetry, $(\mathcal{H}W)(s) - (\mathcal{H}V)(s) \leq \gamma\|V-W\|_\infty$. Hence $|(\mathcal{H}V)(s) - (\mathcal{H}W)(s)| \leq \gamma\|V-W\|_\infty$ for all s, giving (26). □

Theorem 7.5 (Shapley's Theorem [Shapley, 1953]). *Every finite two-player zero-sum discounted stochastic game has a value: there exists a unique $V^* \in \mathbb{R}^{|\mathcal{S}|}$ and stationary strategies π_1^*, π_2^* such that for all $s \in \mathcal{S}$:*

$$V^*(s) = \sup_{\pi_1} \inf_{\pi_2} V_1^{(\pi_1,\pi_2)}(s) = \inf_{\pi_2} \sup_{\pi_1} V_1^{(\pi_1,\pi_2)}(s) = V_1^{(\pi_1^*,\pi_2^*)}(s), \tag{31}$$

and the pair (π_1^*, π_2^*) forms a saddle point: $V_1^{(\pi_1,\pi_2^*)}(s) \leq V^*(s) \leq V_1^{(\pi_1^*,\pi_2)}(s)$ for all π_1, π_2 and all s.

Proof. The proof proceeds in four steps.

Step 1: Existence and uniqueness of the fixed point. By Lemma 7.4, $\mathcal{H}$ is a γ-contraction on the complete metric space $(\mathbb{R}^{|\mathcal{S}|}, \|\cdot\|_\infty)$. By the Banach fixed-point theorem (Lesson 15, Theorem 3.2), there exists a unique $V^* \in \mathbb{R}^{|\mathcal{S}|}$ satisfying $\mathcal{H}V^* = V^*$, i.e., for all s:

$$V^*(s) = \max_{\sigma_1 \in \Delta(\mathcal{A}_1)} \min_{\sigma_2 \in \Delta(\mathcal{A}_2)} \sum_{a_1,a_2} \sigma_1(a_1)\sigma_2(a_2)\big[R(s,a_1,a_2) + \gamma \textstyle\sum_{s'} \mathcal{P}(s'|s,a_1,a_2)V^*(s')\big]. \tag{32}$$

Step 2: Construction of optimal stationary strategies. For each s, the fixed-point equation (32) is a finite two-player zero-sum (matrix) game with payoff matrix $M_s(a_1,a_2) = R(s,a_1,a_2) + \gamma \sum_{s'} \mathcal{P}(s'|s,a_1,a_2)V^*(s')$. By von Neumann's minimax theorem, this matrix game has a value and optimal mixed strategies. Let $\pi_1^*(s) \in \Delta(\mathcal{A}_1)$ and $\pi_2^*(s) \in \Delta(\mathcal{A}_2)$ be the optimal strategies in the matrix game at state s. These choices, made independently at each state, define stationary Markov strategies.

Step 3: V^* equals the value under (π_1^*, π_2^*). Under (π_1^*, π_2^*), the multi-agent Bellman equation (Theorem 6.5) gives:

$$V_1^{(\pi_1^*,\pi_2^*)}(s) = \sum_{a_1,a_2} \pi_1^*(a_1|s)\pi_2^*(a_2|s)\big[R(s,a_1,a_2) + \gamma \textstyle\sum_{s'} \mathcal{P}(s'|s,a_1,a_2)V_1^{(\pi_1^*,\pi_2^*)}(s')\big]. \tag{33}$$

Since $(\pi_1^*(s), \pi_2^*(s))$ achieves the minimax value of the matrix game M_s, the right-hand side equals $(\mathcal{H}V_1^{(\pi_1^*,\pi_2^*)})(s)$. Hence $V_1^{(\pi_1^*,\pi_2^*)}$ is a fixed point of $\mathcal{H}$. By uniqueness, $V_1^{(\pi_1^*,\pi_2^*)} = V^*$.

Step 4: Saddle-point property and minimax equality. We show $V_1^{(\pi_1,\pi_2^*)}(s) \leq V^*(s)$ for all stationary π_1. Define $W = V_1^{(\pi_1,\pi_2^*)}$. By the Bellman equation:

$$W(s) = \sum_{a_1,a_2} \pi_1(a_1|s)\pi_2^*(a_2|s)\big[R(s,a_1,a_2) + \gamma \textstyle\sum_{s'} \mathcal{P}(s'|s,a_1,a_2)W(s')\big] \tag{34}$$

$$\leq \max_{\sigma_1} \min_{\sigma_2} \sum_{a_1,a_2} \sigma_1(a_1)\sigma_2(a_2)\big[R(s,a_1,a_2) + \gamma \textstyle\sum_{s'} \mathcal{P}(s'|s,a_1,a_2)W(s')\big] \tag{35}$$

$$= (\mathcal{H}W)(s). \tag{36}$$

The inequality holds because $\pi_2^*(s)$ is the minimizer in the matrix game with payoff matrix $M_s(V^*)$, and since W may differ from V^*, we use the general inequality $\min_{\sigma_2} f(\sigma_1, \sigma_2) \leq f(\sigma_1, \pi_2^*)$; more precisely, for any fixed π_1:

$$\sum_{a_1,a_2} \pi_1(a_1|s)\pi_2^*(a_2|s)M_s(a_1, a_2; W) \leq \max_{\sigma_1} \min_{\sigma_2} \sum_{a_1,a_2} \sigma_1(a_1)\sigma_2(a_2)M_s(a_1, a_2; W) = (\mathcal{H}W)(s).$$

So $W \leq \mathcal{H}W$ componentwise. Iterating: $W \leq \mathcal{H}W \leq \mathcal{H}^2W \leq \cdots$. Since $\mathcal{H}^kW \to V^*$ (as $\mathcal{H}$ is a contraction with fixed point V^*), we obtain $W \leq V^*$. By a symmetric argument, $V_1^{(\pi_1^*,\pi_2)} \geq V^*$ for all π_2.

Combining, for any π_1, π_2:

$$V_1^{(\pi_1,\pi_2^*)}(s) \leq V^*(s) \leq V_1^{(\pi_1^*,\pi_2)}(s), \tag{37}$$

which is the saddle-point condition. Taking sup over π_1 on the left and inf over π_2 on the right:

$$\sup_{\pi_1} V_1^{(\pi_1,\pi_2^*)}(s) \leq V^*(s) \leq \inf_{\pi_2} V_1^{(\pi_1^*,\pi_2)}(s). \tag{38}$$

But $\sup_{\pi_1} \inf_{\pi_2} V_1^{(\pi_1,\pi_2)}(s) \leq \inf_{\pi_2} \sup_{\pi_1} V_1^{(\pi_1,\pi_2)}(s)$ always holds (weak duality), and the above shows both are equal to $V^*(s)$. This completes the proof. □

Shapley's Theorem (1953)

Every finite two-player zero-sum discounted stochastic game has a unique value $V^* = \mathcal{H}V^*$ (the fixed point of the Shapley operator), and both players have optimal stationary strategies. The proof uses only two ingredients: the minimax theorem for matrix games (to define $\mathcal{H}$) and the Banach fixed-point theorem (to establish the unique fixed point). Value iteration $V^{k+1} = \mathcal{H}V^k$ converges at rate $\|V^k - V^*\|_\infty \leq \gamma^k \|V^0 - V^*\|_\infty$.

Example 7.6 (Value Iteration for a Zero-Sum Stochastic Game)**.** Consider a two-state zero-sum stochastic game with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A}_1 = \mathcal{A}_2 = \{a, b\}$, and $\gamma = 0.9$. At state s_1, the payoff matrix for Player 1 is:

$$R(s_1, \cdot, \cdot) = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix},$$

and all transitions from s_1 go to s_2 deterministically. At s_2, the payoff matrix is:

$$R(s_2, \cdot, \cdot) = \begin{pmatrix} -1 & 3 \\ 1 & 0 \end{pmatrix},$$

with all transitions returning to s_1. Initialize $V^0 = (0, 0)^\top$. The first iteration computes:

$$(\mathcal{H}V^0)(s_1) = \text{val}\begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} = \frac{2}{4} = 0.5,$$
$$(\mathcal{H}V^0)(s_2) = \text{val}\begin{pmatrix} -1 & 3 \\ 1 & 0 \end{pmatrix} = \frac{1}{5} = 0.2,$$

where val$(\cdot)$ denotes the minimax value of the matrix game (computable by linear programming or the 2×2 formula). For the second iteration, the effective payoff matrices include the continuation values: $M_{s_1} = R(s_1, \cdot, \cdot) + 0.9 \cdot V^1(s_2) \cdot \mathbf{1}\mathbf{1}^\top$ (since all transitions from s_1 go to s_2), and similarly for s_2. The iteration converges geometrically at rate 0.9.

Minimax-Q Learning

Minimax-Q learning [Filar and Vrieze, 1997] extends Q-learning to zero-sum stochastic games. The agent maintains $Q(s, a_1, a_2)$ and updates:

$$Q(s, a_1, a_2) \leftarrow (1-\alpha)Q(s, a_1, a_2) + \alpha\big[r + \gamma \cdot \mathrm{val}_{a_1', a_2'} Q(s', a_1', a_2')\big],$$

where val solves the matrix game at s'. Convergence follows from the contraction property of the Shapley operator established in Lemma 7.4, combined with stochastic approximation theory (Lesson 19).

8 General-Sum Stochastic Games

In the general-sum case, players' rewards need not sum to zero, and the minimax framework no longer applies. The appropriate solution concept is the Markov perfect equilibrium.

Definition 8.1 (Markov Perfect Equilibrium)**.** A stationary joint Markov strategy profile $\boldsymbol{\pi}^* = (\pi_1^*, \ldots, \pi_n^*)$ is a *Markov perfect equilibrium* (MPE) if for every player i, every state $s \in \mathcal{S}$, and every stationary Markov strategy π_i for player i:

$$V_i^{(\pi_i^*, \pi_{-i}^*)}(s) \geq V_i^{(\pi_i, \pi_{-i}^*)}(s). \tag{39}$$

That is, no player can improve their value at *any* state by unilaterally deviating to a different stationary strategy, given that all other players follow $\boldsymbol{\pi}^*$.

Remark 8.2. An MPE is a Nash equilibrium in the space of stationary Markov strategies with the additional requirement that it is an equilibrium at every state (not just the initial one). This is the stochastic game analogue of subgame perfect equilibrium in extensive-form games (Lesson 20).

Theorem 8.3 (Fink–Takahashi Theorem [Fink, 1964, Takahashi, 1964])**.** *Every finite n-player discounted stochastic game has at least one Markov perfect equilibrium in stationary (possibly mixed) strategies.*

Proof sketch. The proof generalizes Nash's fixed-point argument to the dynamic setting. We outline the key steps.

Step 1: Best-response value operator. Fix the strategies of all players except i at π_{-i}. Player i then faces a single-agent MDP with transition kernel $\mathcal{P}^{\pi_{-i}}(s'|s, a_i) = \sum_{a_{-i}} \prod_{j \neq i} \pi_j(a_j|s)\mathcal{P}(s'|s, a_i, a_{-i})$ and reward $R_i^{\pi_{-i}}(s, a_i) = \sum_{a_{-i}} \prod_{j \neq i} \pi_j(a_j|s) R_i(s, a_i, a_{-i})$. By standard MDP theory (Lesson 45), this MDP has an optimal stationary policy, and the optimal value satisfies a contraction fixed-point equation. Define the best-response correspondence $\mathrm{BR}_i(\pi_{-i})$ as the set of optimal stationary policies for this induced MDP.

Step 2: Joint best-response correspondence. Let $\Pi = \prod_{i=1}^n \prod_{s \in \mathcal{S}} \Delta(\mathcal{A}_i)$ be the space of all stationary joint Markov strategy profiles. This is a nonempty, compact, convex subset of a finite-dimensional Euclidean space. Define $\mathrm{BR}(\boldsymbol{\pi}) = \prod_{i=1}^n \mathrm{BR}_i(\pi_{-i})$.

Step 3: Verify Kakutani hypotheses. One must show that $\mathrm{BR}(\boldsymbol{\pi})$ is nonempty and convex for each $\boldsymbol{\pi}$, and that BR has a closed graph. Nonemptiness follows from the existence of optimal MDP policies. Convexity of $\mathrm{BR}_i(\pi_{-i})$ requires care: while the set of optimal *values* is a singleton (by contraction), the set of optimal *policies* can be multi-valued but is convex when we consider the set of policies achieving the maximum at each state independently (since the optimality condition is linear in π_i at each state, given the value function). The closed graph property follows from the continuity of the value function in the joint strategy profile (which itself follows from the contraction mapping depending continuously on π_{-i}).

Step 4: Apply Kakutani's fixed-point theorem. By Kakutani's theorem, BR has a fixed point $\boldsymbol{\pi}^*$, meaning $\boldsymbol{\pi}^* \in \mathrm{BR}(\boldsymbol{\pi}^*)$. This fixed point satisfies (39) and is therefore an MPE. □

Markov Perfect Equilibrium Existence

Every finite discounted stochastic game possesses at least one Markov perfect equilibrium in stationary mixed strategies. This guarantees that equilibrium analysis of multi-agent sequential interaction is always well-defined. However, MPEs may be non-unique, and the existence proof (via Kakutani's fixed-point theorem) is non-constructive—it does not yield an efficient algorithm for finding an MPE.

Warning

Unlike zero-sum stochastic games (where the value is unique and value iteration converges), general-sum stochastic games may have *multiple* MPEs with different value vectors. There is no canonical way to select among them. Moreover, simple dynamic programming-style iterations (e.g., repeatedly computing stage-game Nash equilibria at each state) are not guaranteed to converge in the general-sum case. This fundamental difficulty pervades all of MARL.

Example 8.4 (General-Sum Two-State Game)**.** Consider a two-player game with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A}_1 = \mathcal{A}_2 = \{a, b\}$, and $\gamma = 0.5$. At s_1, the bimatrix game is:

	a	b
a	$(3,3)$	$(0,4)$
b	$(4,0)$	$(1,1)$

with all transitions going to s_2, and at s_2 the game is:

	a	b
a	$(2,2)$	$(0,0)$
b	$(0,0)$	$(1,1)$

with all transitions returning to s_1. Each state has the structure of a Prisoner's Dilemma or coordination game. The coupling through transitions and discounting can create MPEs that differ from the static Nash equilibria at each state. For instance, cooperative behavior at s_1 may be sustained if the continuation value at s_2 sufficiently rewards cooperation.

9 Equilibrium Computation and Complexity

The existence theorems of the preceding sections guarantee that equilibria exist, but they do not address the computational question: how hard is it to *find* an equilibrium?

9.1 Computing Nash Equilibria in Normal-Form Games

For two-player games (bimatrix games), several algorithms exist.

Definition 9.1 (Support Enumeration)**.** A Nash equilibrium (σ_1^*, σ_2^*) satisfies the *indifference principle*: every action in the support of σ_i^* yields the same expected payoff against σ_{-i}^*. The *support enumeration* method iterates over all possible support pairs (S_1, S_2) with $S_i \subseteq \mathcal{A}_i$ and solves the resulting system of linear equations. The number of support pairs is $\sum_{k_1=1}^{|\mathcal{A}_1|} \sum_{k_2=1}^{|\mathcal{A}_2|} \binom{|\mathcal{A}_1|}{k_1}\binom{|\mathcal{A}_2|}{k_2}$, which is exponential in the action space sizes.

Remark 9.2 (Lemke–Howson Algorithm). The Lemke–Howson algorithm [Shoham and Leyton-Brown, 2008] finds one Nash equilibrium of a bimatrix game by following a path on a labeled polytope (a complementary pivoting method, analogous to the simplex method for linear programming). While it always terminates with a Nash equilibrium, it may require exponentially many pivot steps in the worst case [Shoham and Leyton-Brown, 2008].

9.2 PPAD-Completeness

The computational intractability of Nash equilibrium computation was established in a celebrated sequence of results.

Definition 9.3 (The Class PPAD)**.** PPAD ("Polynomial Parity Arguments on Directed graphs") is the complexity class of total search problems reducible to the End-of-the-Line problem: given an exponentially large directed graph (specified by polynomial-time successor and predecessor circuits) with a known source vertex of in-degree 0, find either another source or a sink (vertex of out-degree 0). Such a vertex is guaranteed to exist by a parity argument.

Theorem 9.4 (PPAD-Completeness of Nash Equilibrium [Daskalakis et al., 2009, Chen et al., 2009])**.** *Computing a Nash equilibrium of a two-player normal-form game (even an approximate Nash equilibrium with inverse polynomial precision) is PPAD-complete.*

Remark 9.5. The PPAD-completeness result means that, unless PPAD $\subseteq$ P (which is considered unlikely and would have far-reaching consequences), there is no polynomial-time algorithm for computing Nash equilibria. This is in stark contrast to zero-sum games, where the minimax equilibrium can be computed in polynomial time via linear programming.

9.3 Implications for Stochastic Games

Proposition 9.6. *Computing a Markov perfect equilibrium of a general-sum stochastic game is at least as hard as computing a Nash equilibrium of a normal-form game, and is therefore PPAD-hard.*

Proof. Any normal-form game G can be embedded as a stochastic game with a single state ($|\mathcal{S}| = 1$) and $\gamma = 0$. An MPE of this stochastic game is precisely a Nash equilibrium of G. Since computing a Nash equilibrium of a bimatrix game is PPAD-complete (Theorem 9.4), computing an MPE of a stochastic game is PPAD-hard. □

Remark 9.7 (Approximate Equilibria). Given the computational barriers, practical MARL algorithms typically seek *approximate* equilibria. A strategy profile $\boldsymbol{\pi}$ is an *ε-Nash equilibrium* if no player can improve their value by more than ε through unilateral deviation. Even computing ε-Nash equilibria with $\varepsilon = 1/\text{poly}(n)$ remains PPAD-hard for general games. However, for specific game structures (zero-sum, potential, polymatrix), more efficient algorithms are available.

Computational Complexity and MARL Algorithm Design

The PPAD-hardness of computing Nash equilibria fundamentally shapes MARL algorithm design. Algorithms like Nash-Q, which require solving a Nash equilibrium at every update step, are impractical for large action spaces. This motivates:

- **Zero-sum specialization:** Restricting to zero-sum games where linear programming suffices (Minimax-Q).
- **No-regret approaches:** Using independent no-regret learners to converge to CCE (Section 5).
- **Policy gradient methods:** MADDPG and MAPPO bypass equilibrium computation entirely by performing gradient ascent on each agent's objective.

- **Self-play:** In symmetric zero-sum games, self-play with best-response dynamics converges to Nash equilibrium.

10 Exercises

Exercise 10.1 (Nash Equilibrium in a 2×2 Game)**.** $(*)$
Consider the two-player game with payoff matrices:

	L	R
U	$(4, 2)$	$(1, 3)$
D	$(2, 4)$	$(3, 1)$

Find all Nash equilibria (pure and mixed). For the mixed equilibrium, compute the expected payoff to each player.

Hint: Use the indifference principle: in a completely mixed equilibrium, each player must be indifferent among all actions in the support.

Exercise 10.2 (Bellman Equations for a Two-State Stochastic Game)**.** $(*)$
Consider the zero-sum stochastic game from Example 7.6. Write down the Bellman fixed-point equation $V^* = \mathcal{H}V^*$ explicitly for both states. Perform three iterations of value iteration starting from $V^0 = (0, 0)^\top$ and verify that the iterates converge. Compute the minimax strategies at each state for the final iterate.

Exercise 10.3 (Regret Bound for Exp3)**.** $(**)$
The Exp3 algorithm is a bandit variant of multiplicative weights where player i only observes the payoff of the chosen action (not the full payoff vector). The estimated payoff is $\hat{u}_i^t(a_i) = u_i(a_i, a_{-i}^t)\mathbf{1}[a_i^t = a_i]/\sigma_i^t(a_i)$. Show that $\mathbb{E}[\hat{u}_i^t(a_i)] = u_i(a_i, a_{-i}^t)$ (unbiasedness) and $\mathbb{E}[\hat{u}_i^t(a_i)^2] \leq 1/\sigma_i^t(a_i)$ (bounded variance). Using these properties and the MWU potential argument from Theorem 5.3, prove that Exp3 achieves expected regret $O(\sqrt{T|\mathcal{A}_i| \ln |\mathcal{A}_i|})$.

Exercise 10.4 (Shapley Operator Contraction Details)**.** $(**)$
In the proof of Lemma 7.4, we used the fact that $|\mathrm{val}(A) - \mathrm{val}(B)| \leq \|A - B\|_\infty$ for any two matrices A, B of the same size, where $\mathrm{val}(\cdot)$ denotes the minimax value and $\|A - B\|_\infty = \max_{i,j} |A_{ij} - B_{ij}|$. Prove this fact directly from the definition of the minimax value, without invoking LP duality.

Hint: Let (σ_1^*, σ_2^*) achieve $\mathrm{val}(A)$ and consider $\sigma_1^{*\top} B \sigma_2^*$.

Exercise 10.5 (Average-Reward Stochastic Games)**.** $(***)$
In an average-reward (undiscounted) zero-sum stochastic game, the value to Player 1 starting from s is $V(s) = \liminf_{T\to\infty} \frac{1}{T}\mathbb{E}[\sum_{t=0}^{T-1} R(s^t, \mathbf{a}^t)|s^0 = s]$.

(a) Explain why the Shapley operator is no longer a strict contraction when $\gamma = 1$. What goes wrong with the Banach fixed-point argument?

(b) Under the assumption that the stochastic game is *irreducible* (every stationary joint strategy induces an irreducible Markov chain), state the analogue of the Bellman optimality equation using the gain-bias framework (cf. Lesson 44 for the single-agent case).

(c) Discuss whether the existence of a value is still guaranteed. Cite relevant results from Filar and Vrieze [1997].

Exercise 10.6 (Shapley's Counterexample for Fictitious Play)**.** $(***)$
Consider the following symmetric 3×3 game (Shapley, 1964):

	R	P	S
R	$(0,0)$	$(1,0)$	$(0,1)$
P	$(0,1)$	$(0,0)$	$(1,0)$
S	$(1,0)$	$(0,1)$	$(0,0)$

This is a variant of Rock-Paper-Scissors.

(a) Find the unique Nash equilibrium of this game.

(b) Simulate fictitious play starting from (R, R) for 20 rounds and show that the empirical frequencies cycle rather than converge.

(c) Prove that the empirical frequencies in this game do not converge to the Nash equilibrium under fictitious play (or provide the key argument for the cyclic behavior).

Hint: Track which pure best responses are active and show the sequence of best responses cycles through a pattern that prevents the empirical frequencies from settling.

11 Summary

- A **normal-form game** $(\mathcal{N}, \{\mathcal{A}_i\}, \{u_i\})$ models simultaneous strategic interaction; Nash's theorem guarantees existence of mixed-strategy equilibria via Kakutani's fixed-point theorem.
- In **repeated games**, the folk theorem shows that patient players can sustain a vast set of equilibrium payoffs beyond the static Nash equilibria, using punishment strategies.
- **Fictitious play** converges to Nash equilibrium in two-player zero-sum games (Robinson's theorem) but may fail to converge in general games (Shapley's counterexample).
- **No-regret learning** (e.g., multiplicative weights) achieves sublinear regret $O(\sqrt{T \ln |\mathcal{A}_i|})$; when all players use no-regret algorithms, the empirical joint play converges to a coarse correlated equilibrium.
- A **stochastic game** $(\mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \mathcal{P}, \{R_i\}, \gamma)$ generalizes both MDPs (single player) and repeated games (single state), providing the foundational model for multi-agent RL.
- The **multi-agent Bellman equation** extends the single-agent version with a product of independent strategy probabilities, coupling each player's value to the joint strategy profile.
- **Shapley's theorem**: zero-sum discounted stochastic games have a unique value and optimal stationary strategies, proved via the Banach fixed-point theorem applied to the γ-contractive Shapley operator.
- **Fink–Takahashi theorem**: general-sum discounted stochastic games have at least one Markov perfect equilibrium in stationary strategies, proved via Kakutani's fixed-point theorem on the joint best-response correspondence.
- **Computational complexity**: computing Nash equilibria (even in two-player normal-form games) is PPAD-complete, and computing MPEs in stochastic games is at least PPAD-hard. This hardness motivates approximate and structure-exploiting algorithms in MARL.
- The **zero-sum vs. general-sum divide** is fundamental: zero-sum games admit unique values, LP-based computation, and convergent learning algorithms, while general-sum games face multiplicity of equilibria and computational intractability.

12 Further Reading

- **Lesson 62** (Cooperative MARL) extends the stochastic game framework to cooperative settings, introducing Dec-POMDPs, centralized training with decentralized execution (CTDE), and value decomposition methods (VDN, QMIX, QTRAN).
- **Lesson 63** (Competitive and Mixed MARL) develops algorithms for competitive stochastic games, including Minimax-Q, Nash-Q, MADDPG, and self-play methods such as PSRO.
- **Lesson 64** (Mean-Field Games) takes the infinite-player limit of stochastic games via McKean–Vlasov dynamics, providing tractable approximations for large populations.
- **Lesson 65** (Communication in MARL) studies emergent communication protocols (CommNet, DIAL, TarMAC) within the stochastic game framework.
- Filar and Vrieze [1997] provides the most complete mathematical treatment of competitive Markov decision processes, including average-reward stochastic games, ordered field properties, and algebraic approaches to equilibrium computation.
- Cesa-Bianchi and Lugosi [2006] offers a comprehensive treatment of prediction, learning, and games, with detailed analysis of regret minimization, calibration, and connections to game theory.
- Hart and Mas-Colell [2000] develops the theory of simple adaptive procedures leading to correlated equilibrium, extending the no-regret framework to stronger solution concepts.
- For the original proof that zero-sum games have a value, see von Neumann [1928]; Shapley's extension to stochastic games [Shapley, 1953] is remarkably concise and accessible.
- Zhang et al. [2021] provides a modern survey connecting the classical stochastic game theory of this lesson to contemporary deep MARL algorithms.

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